

Interface Capability Brief

EDI 270/271 Realtime Eligibility · Stakeholder overview · 2026-08-13

1. Executive summary

Clinic (or EHR) posts once. PilotFish owns the full synchronous 270 → payer → 271 round-trip and returns clinic-ready JSON on the same HTTP request.

2. Who this is for

- Sources: Clinic Web UI posts EligibilityRequest XML to PilotFish /eip/rest/eligibility/check - Destinations: mock payer X12 271; clinic JSON summary; audit files under output/ - Demo vs production: Demo only — canned AAA/EB, no SNIP, no clearinghouse

- Sources: Clinic Web UI posts EligibilityRequest XML to PilotFish /eip/rest/eligibility/check
- Destinations: mock payer X12 271; clinic JSON summary; audit files under output/
- Demo vs production: Demo only — canned AAA/EB, no SNIP, no clearinghouse

3. What the interface can do

Capabilities below are drawn from the working design and, when present, the published CapabilityStatement — not from aspirational roadmap items.

- Transport: RESTful Web Service (SERVICE_NAME=eligibility, resource check, Synchronous=true)
- Format: XML request → PilotFish EDI XML → X12 270 → HttpPost → wrap 271 in <EdiPayload> → JSON
- Identity fields: MemberId (NM1IL...MI), TraceNumber (TRN02 / BHT03)
- Samples: TableData examples in samples/; live story uses FAIL001 / OK001 / UNKNOWN

Integration highlights

FQCN sources: CRL Plus REST, edi-270-271-eligibility EDI/XSLT, FHIR Keycloak HttpPost + PostProcessors + SynchronousResponseProcessor. Image: pilotfish-eip:23R1.

Stage	Module / mechanism	Notes
Listener	RESTfulWebServiceListener	Skeleton: CRL Plus Status.91 REST listener. Synchronous=true because this is request/response.
Router	XPathRoutingModule OGNL on HttpMethodName + ResourceName	check vs 405
Build	XSLT → XML Formatting → EDITransformationProcessor XML→EDI	TableData 270-A1 / 271-A1. XML Formatting on the 270 EDI XML audit write.
Payer	HttpPostTransport	Skeleton: FHIR Route 0 - Keycloak JWT Auth. Target \$\$PAYER_X12_URL.
Parse	Save body → wrap <EdiPayload> → structural Saxon XSLT	Same 23R1 honesty as orig: no EDI→XML 271.
Sync reply	SynchronousResponseProcessor in PostProcessors	Continues after HttpPost (Keycloak pattern).

4. Honest boundaries (not claimed yet)

-
- Switch parse to EDITransformationProcessor EDI→XML once 271 TableData is proven on 23R1?

5. What you can verify in a walkthrough

Use these as checklist items when reviewing the demo with business stakeholders. They describe outcomes, not module names.

- Exercise route “1 - Realtime Eligibility Check” from the Web UI or documented smoke path
- Confirm outputs land in the documented folders / queues and appear in the Web UI status views

6. How it works (plain language)

7. Validation

- What is checked: 271 has AAA or EB; clinic banner from parse JSON; UI mode=realtime - What is NOT checked: SNIP 270/271, real payer connectivity, HIPAA code-set licensing - Does failure block outbound? yes for unparseable request XML

6. State & idempotency

- Status model: none (sync request/response) - Dedup keys: TraceNumber on the wire - Retry / poison: bad XML fails at XSLT; unknown methods 405

8. Dual-write / side effects

- Order: sync JSON is the contract; file snapshots are audit only - Compensation: none - Demo shortcuts: mock payer matches on NM1IL and does not require ST270; wrap 271 in EdiPayload for the parse XSLT

3. Inbound contract

- Transport: RESTful Web Service (SERVICE_NAME=eligibility, resource check, Synchronous=true) - Format: XML request → PilotFish EDI XML → X12 270 → HttpPost → wrap 271 in <EdiPayload> → JSON - Identity fields: MemberId (NM1IL...MI), TraceNumber (TRN02 / BHT03) - Samples: TableData examples in samples/; live story uses FAIL001 / OK001 / UNKNOWN

Routes in this interface

Each route is a configured PilotFish flow. Technical wiring appears in the Route Diagrams PDF; here is the stakeholder reading.

1 - Realtime Eligibility Check: runs steps such as Realtime Eligibility REST, Conditional Router, Post 270 To Payer, Snapshot Raw 271.

1 - Realtime Eligibility Check: runs steps such as Realtime Eligibility REST, Snapshot Inbound Request, Conditional Router, Post 270 To Payer.

1 - Realtime Eligibility Check: runs steps such as Realtime Eligibility REST, Snapshot Inbound Request, Conditional Router, Map Request To 270 EDI XML.

1 - Realtime Eligibility Check: runs steps such as Realtime Eligibility REST, Snapshot Inbound Request, Conditional Router, Map Request To 270 EDI XML.

1 - Realtime Eligibility Check: runs steps such as Realtime Eligibility REST, Snapshot Inbound Request, Conditional Router, Map Request To 270 EDI XML.

1 - Realtime Eligibility Check: runs steps such as Realtime Eligibility REST, Snapshot Inbound Request, Conditional Router, Map Request To 270 EDI XML.

1 - Realtime Eligibility Check: runs steps such as Realtime Eligibility REST, Snapshot Inbound Request, Conditional Router, Map Request To 270 EDI XML.

1 - Realtime Eligibility Check: runs steps such as Realtime Eligibility REST, Snapshot Inbound Request, Conditional Router, Map Request To 270 EDI XML.

1 - Realtime Eligibility Check: runs steps such as Realtime Eligibility REST, Snapshot Inbound Request, Conditional Router, Map Request To 270 EDI XML.

1 - Realtime Eligibility Check: runs steps such as Realtime Eligibility REST, Snapshot Inbound Request, Conditional Router, Map Request To 270 EDI XML.

1 - Realtime Eligibility Check: runs steps such as Realtime Eligibility REST, Snapshot Inbound Request, Conditional Router, Map Request To 270 EDI XML.

1 - Realtime Eligibility Check: runs steps such as Realtime Eligibility REST, Snapshot Inbound Request, Conditional Router, Map Request To 270 EDI XML.

7. Security & trust posture

- OAuth2 / OIDC Bearer tokens (Keycloak in this demo) protect write and export operations
- X12 SNIP validation is used where configured to reject bad EDI before downstream work
- Credentials and passwords in this package are demo-only — not production secrets
- Risk noted in design: High — Mock payer is not a real 271 engine — Pitch is eligibility theater
- Risk noted in design: Med — 23R1 trial X12 tables — XML→EDI may omit ST; EDI→XML 271 unused
- Risk noted in design: Med — Duplicate processor names — EIP will not load the route
- Risk noted in design: Low — Clinic UI still builds EligibilityRequest XML — Production could POST the same XML from any clinic system

8. Seeing it work

- - Ports: Mock payer 8211, EIP 8120, Web UI 8121 - Volumes: ./output, ./logs, ./samples, ./documents, TableData mount (5 ../ from Insurance/EDI/) - Heap: 512M–2GB - Dependencies / cold start: EIP ~60–90s - Compose project: edi-270-271-realtime
- Local: `http://127.0.0.1:8121/`
- EIP: `http://127.0.0.1:8120/eip/`
- Eligibility check: `http://127.0.0.1:8120/eip/rest/eligibility/check`
- Mock payer: `http://127.0.0.1:8211/x12/270`
- `docker compose --profile full up -d --build`

9. Related technical documents

- DESIGN.md — working engineering specification for this interface
- README.md — how to run, ports, and smoke commands
- This Capability Brief — shareable stakeholder summary (regenerate after design changes)

This document is generated from the interface sources listed above. If design and runtime diverge, believe the smoke-tested runtime and update DESIGN.md.